TCS NQT

CODING SECTION

Exam-Style Question Bank · 2025–2026 Edition

Ninja · Digital · Prime

37 Questions · Full Exam Format · Constraints · Sample I/O · Hints

Section A	Section B	Time Limit
Questions 1 – 25	Questions 26 – 37	90 min (Shared)
Confirmed from Previous Papers	Predicted High-Probability Topics	For both coding questions

SECTION A — CONFIRMED QUESTIONS

Questions 1 to 25 · Directly reported from TCS NQT previous year papers (2022 – 2025)

Q1	Chocolate Packets — Move Zeros to End Section A · Arrays / Two Pointer	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Two Pointer	100 Points	90 min (Shared for both questions)

Problem Scenario

Ravi works in a chocolate factory. Some packets on the conveyor belt are empty (marked as 0). The manager wants all empty packets moved to the end of the belt without changing the order of the filled packets.

Problem Statement

Given an array of N integers (some may be 0), rearrange it so all zeros appear at the end while preserving the original relative order of all non-zero elements. The operation must be performed in-place without allocating extra space.

Constraints

- `1 <= N <= 100000`
- `0 <= arr[i] <= 10000`

Input Format

Line 1: integer N

Line 2: N space-separated integers

Output Format

Print the rearranged array as N space-separated integers on a single line.

Sample Test Cases

Sample Input 1	Sample Output 1
6	4 5 1 9 0 0

4 0 5 0 1 9	

Explanation: Non-zero elements 4,5,1,9 preserve their order; both zeros move to the end.

Sample Input 2	Sample Output 2
5 0 0 0 1 2	1 2 0 0 0

Explanation: Three leading zeros shift to the end after all non-zero elements.

Approach / Hint

Two-pointer technique: 'insertPos' starts at 0. Scan array with 'i'; whenever arr[i] != 0, place arr[i] at insertPos and increment insertPos. After the scan, fill remaining positions with 0.

Q2	Second Largest Element Section A · Arrays / Linear Scan	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Linear Scan	100 Points	90 min (Shared for both questions)

Problem Scenario

Priya has recorded scores of N students. She needs the second highest score for a merit certificate (the top scorer already received the first prize).

Problem Statement

Given an array of N integers, find the second largest distinct element. If no second largest exists (all elements are equal or only one distinct value exists), print -1.

Constraints

- `2 <= N <= 100000`
- `1 <= arr[i] <= 1000000`

Input Format

Line 1: integer N

Line 2: N space-separated integers

Output Format

Print the second largest element, or -1 if it does not exist.

Sample Test Cases

Sample Input 1	Sample Output 1
5 10 5 20 8 20	10

Explanation: Largest is 20; the second largest distinct value is 10.

Sample Input 2	Sample Output 2
4 7 7 7 7	-1

Explanation: All elements are equal — no second largest exists.

Approach / Hint

Maintain two variables: largest and secondLargest (both initialised to -INF). Traverse array once. If arr[i] > largest: secondLargest = largest, largest = arr[i]. Else if arr[i] > secondLargest and arr[i] != largest: secondLargest = arr[i]. O(n) time, O(1) space — do NOT sort.

Q3	Sort 0s, 1s and 2s — Color Flag Section A · Arrays / Dutch National Flag	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Dutch National Flag	100 Points	90 min (Shared for both questions)

Problem Scenario

A gardener has a row of flowers that are either red (0), white (1), or blue (2). She wants them sorted by color — all red first, then white, then blue — in a single pass without using any sorting library.

Problem Statement

Given an array containing only the values 0, 1, and 2, sort it in ascending order in O(n) time and O(1) extra space. You must NOT use any built-in sort function.

Constraints

- `1 <= N <= 100000`
- `arr[i] is 0, 1, or 2 only`

Input Format

Line 1: integer N

Line 2: N space-separated integers (each 0, 1, or 2)

Output Format

Print the sorted array as N space-separated integers.

Sample Test Cases

Sample Input 1	Sample Output 1
7 2 0 1 1 0 2 1	0 0 1 1 1 2 2

Explanation: All 0s first, then 1s, then 2s.

Sample Input 2	Sample Output 2
4 1 2 0 1	0 1 1 2

Approach / Hint

Dutch National Flag algorithm: three pointers — low=0, mid=0, high=N-1. While mid<=high: if arr[mid]==0 swap(arr[low],arr[mid]), low++, mid++; if arr[mid]==1 mid++; if arr[mid]==2 swap(arr[mid],arr[high]), high--.

Q4	Missing Roll Number Section A · Arrays / Math	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Math	100 Points	90 min (Shared for both questions)

Problem Scenario

A teacher has roll numbers 1 to N for a class of N students. During attendance one roll number is missing from the register. Find it quickly without sorting.

Problem Statement

Given an array of N-1 distinct integers, each in the range [1, N], find the single missing integer.

Constraints

- `2 <= N <= 100000`
- `1 <= arr[i] <= N`
- `All values are distinct`

Input Format

Line 1: integer N

Line 2: N-1 space-separated integers

Output Format

Print the single missing integer on one line.

Sample Test Cases

Sample Input 1	Sample Output 1
5 1 2 4 5	3

Explanation: Expected sum = 1+2+3+4+5 = 15; actual sum = 1+2+4+5 = 12; missing = 15 - 12 = 3.

Sample Input 2	Sample Output 2
6 6 3 4 1 5	2

Approach / Hint

Formula approach: Missing = N*(N+1)/2 - sum(array). No sorting, no HashMap needed. O(n) time, O(1) space.

Q5	Rotate the Regiment — Array Rotation Section A · Arrays / Reversal Algorithm	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Reversal Algorithm	100 Points	90 min (Shared for both questions)

Problem Scenario

A commander orders his regiment of N soldiers to rotate their positions K steps to the right. Help him visualise the new formation.

Problem Statement

Given an array of N integers and a value K, rotate the array to the right by K positions. If K >= N, use effective rotation K = K % N.

Constraints

- `1 <= N <= 100000`
- `0 <= K <= 100000`
- `-10000 <= arr[i] <= 10000`

Input Format

Line 1: two integers N and K separated by a space

Line 2: N space-separated integers

Output Format

Print the rotated array as N space-separated integers.

Sample Test Cases

Sample Input 1	Sample Output 1
7 3 1 2 3 4 5 6 7	5 6 7 1 2 3 4

Explanation: After 3 right rotations the last 3 elements come to the front.

Sample Input 2	Sample Output 2
5 7	4 5 1 2 3

1 2 3 4 5	

Explanation: K=7; effective rotation = 7 % 5 = 2.

Approach / Hint

Reversal method (O(n) time, O(1) space): Step 1 — reverse entire array. Step 2 — reverse first K elements. Step 3 — reverse remaining N-K elements.

Q6	Split Array with Equal Averages Section A · Arrays / Prefix Sum	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Prefix Sum	100 Points	90 min (Shared for both questions)

Problem Scenario

A logistics manager wants to split a shipment into two consecutive groups so that both groups have the same average weight. Determine if such a split point exists.

Problem Statement

Given an array of N integers, determine whether there exists an index i (1 <= i <= N-1) such that the average of arr[0..i-1] equals the average of arr[i..N-1]. Print YES or NO.

Constraints

- `2 <= N <= 100000`
- `1 <= arr[i] <= 10000`

Input Format

Line 1: integer N

Line 2: N space-separated integers

Output Format

Print YES if a valid split exists, otherwise NO.

Sample Test Cases

Sample Input 1	Sample Output 1
5 3 1 2 6 3	YES

Explanation: Split at index 4: left=[3,1,2,6] avg=3.0; right=[3] avg=3.0. Equal!

Sample Input 2	Sample Output 2
3 1 2 4	NO

Explanation: No valid split point gives equal averages.

Approach / Hint

For each split point i (1 to N-1): leftSum = prefix[i], rightSum = totalSum - leftSum. Check leftSum*(N-i) == rightSum*i (cross-multiply to avoid float division). Track leftSum incrementally.

Q7	Hotel Guest Counter — Peak Occupancy Section A · Arrays / Greedy / Sorting	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Greedy / Sorting	100 Points	90 min (Shared for both questions)

Problem Scenario

A hotel manager has check-in and check-out records of guests and wants to know the peak occupancy — the maximum number of guests ever present simultaneously.

Problem Statement

Given two arrays entry[] and exit[] of size N (integer times), find the maximum number of guests present at any single moment. A guest at time T is present if entry[i] <= T <= exit[i].

Constraints

- `1 <= N <= 100000`
- `1 <= entry[i] <= exit[i] <= 100000`

Input Format

Line 1: integer N

Line 2: N space-separated entry times

Line 3: N space-separated exit times

Output Format

Print a single integer — the maximum simultaneous guest count.

Sample Test Cases

Sample Input 1	Sample Output 1
3 1 2 9 5 5 12	2

Explanation: At times 2 to 5 guests 1 and 2 are both present simultaneously. Max = 2.

Sample Input 2	Sample Output 2

4 1 3 5 7 2 4 6 8	1

Explanation: No two guests overlap.

Approach / Hint

Sort both arrays. Two-pointer scan: i=0, j=0, count=0, max=0. While i<N: if entry[i] <= exit[j] then count++, max=max(max,count), i++; else count--, j++. Return max.

Q8	Palindrome Password Check Section A · Strings / Two Pointer	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Strings / Two Pointer	100 Points	90 min (Shared for both questions)

Problem Scenario

A cybersecurity tool must flag passwords that read the same forwards and backwards — palindrome passwords are trivially guessable and must be rejected.

Problem Statement

Given a string S, determine whether it is a palindrome (case-insensitive). Print YES if it is a palindrome, else NO.

Constraints

- `1 <= |S| <= 100000`
- `S contains only English alphabets (upper and/or lower case)`

Input Format

A single line containing the string S

Output Format

Print YES or NO on a single line.

Sample Test Cases

Sample Input 1	Sample Output 1
`RaceCar`	`YES`

Explanation: Lowercased: 'racecar' — reads the same forwards and backwards.

Sample Input 2	Sample Output 2
`Hello`	`NO`

Explanation: 'hello' reversed is 'olleh', which differs.

Approach / Hint

> Two-pointer: left=0, right=|S|-1. Compare S[left].lower() with S[right].lower(). If any mismatch found, return NO. If loop completes without mismatch, return YES. O(n) time.

Q9	Anagram Detector Section A · Strings / Frequency Map	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Strings / Frequency Map	100 Points	90 min (Shared for both questions)

Problem Scenario

Two secret agents use anagram code words. Given two words, determine if one is a rearrangement of the other — meaning they use exactly the same letters with the same frequencies.

Problem Statement

Given two strings A and B of the same length (all lowercase), check if A is an anagram of B. Print YES or NO. NOTE: Do NOT sort the strings — use a frequency map for an O(n) solution.

Constraints

- `1 <= |A| = |B| <= 100000`
- `Both strings contain only lowercase English letters`

Input Format

Line 1: string A

Line 2: string B

Output Format

Print YES or NO on a single line.

Sample Test Cases

Sample Input 1	Sample Output 1
listen silent	YES

Explanation: Both contain the letters e, i, l, n, s, t — same count.

Sample Input 2	Sample Output 2
hello world	NO

Explanation: Different character frequencies.

Approach / Hint

Create an integer array freq[26] initialised to 0. For each character in A: freq[c-'a']++. For each character in B: freq[c-'a']--. If all values in freq are 0, output YES; otherwise NO.

Q10	First Non-Repeating Character Section A · Strings / HashMap	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Strings / HashMap	100 Points	90 min (Shared for both questions)

Problem Scenario

A data analyst processes log entries and needs the first unique character in a log string to serve as a transaction key. Help her find it quickly.

Problem Statement

Given a string S of lowercase English letters, find and print the first character that does not repeat anywhere in S. If every character repeats, print -1.

Constraints

- `1 <= |S| <= 100000`
- `S contains only lowercase English letters`

Input Format

A single line containing the string S

Output Format

Print the first non-repeating character, or -1 if none exists.

Sample Test Cases

Sample Input 1	Sample Output 1
`aabbcde`	`c`

Explanation: 'a' and 'b' both repeat. 'c' appears exactly once and appears before 'd' and 'e'.

Sample Input 2	Sample Output 2
`aabb`	`-1`

Explanation: All characters repeat — no unique character exists.

Approach / Hint

Pass 1: build a HashMap of character frequencies (or int[26] array). Pass 2: traverse string again; return the first character whose frequency equals 1.

Q11	Curtain Color Chunks — Max 'a' Count Section A · Strings / Substring	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Strings / Substring	100 Points	90 min (Shared for both questions)

Problem Scenario

An interior designer considers one floor at a time (a chunk of L rooms). Each room is either aqua ('a') or beige ('b'). Find the floor with the most aqua rooms.

Problem Statement

Given a string S of characters 'a' and 'b', and an integer L, divide S into consecutive chunks of length L (the last chunk may be shorter than L). Print the maximum count of 'a' across all chunks.

Constraints

- `1 <= L <= |S| <= 100000`
- `S contains only 'a' and 'b'`

Input Format

Line 1: string S

Line 2: integer L

Output Format

Print the maximum count of 'a' in any single chunk.

Sample Test Cases

Sample Input 1	Sample Output 1
aabbaab 2	2

Explanation: Chunks: 'aa'(2), 'bb'(0), 'aa'(2), 'b'(0). Maximum = 2.

Sample Input 2	Sample Output 2
aaabba 3	3

Explanation: Chunks: 'aaa'(3), 'bba'(1). Maximum = 3.

Approach / Hint

Loop with step L: for i = 0, L, 2L, ... take slice S[i : i+L], count 'a' in it, update a running maximum. O(n) total.

Q12	Reverse the Sentence Section A · Strings / Manipulation	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Strings / Manipulation	100 Points	90 min (Shared for both questions)

Problem Scenario

A telegram transmission reverses the order of words in a message. Write a decoder that restores the original word order while keeping each word's characters intact.

Problem Statement

Given a sentence of words separated by single spaces (no leading/trailing spaces), reverse the order of the words. Each word's internal characters remain unchanged.

Constraints

- `1 <= |S| <= 10000`
- `S contains only lowercase letters and single spaces`
- `No leading or trailing spaces`

Input Format

A single line containing the sentence

Output Format

Print the sentence with word order reversed on a single line.

Sample Test Cases

Sample Input 1	Sample Output 1
`hello world how are you`	`you are how world hello`

Explanation: Word order is fully reversed.

Sample Input 2	Sample Output 2
`tcs nqt exam`	`exam nqt tcs`

Approach / Hint

Split string on spaces to get a list of words. Reverse the list. Join with a single space and print. In Python: print(' '.join(input().split()[::-1]))

Q13	Count Palindrome Numbers in Range Section A · Strings / Number Conversion	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Strings / Number Conversion	100 Points	90 min (Shared for both questions)

Problem Scenario

A mathematician calls a number 'magical' if it reads the same forwards and backwards. Given a range [M, N], count all magical numbers within it.

Problem Statement

Given two integers M and N (M <= N), count the total number of palindrome integers in the closed range [M, N].

Constraints

- `0 <= M <= N <= 10000`

Input Format

A single line with two space-separated integers M and N

Output Format

Print the count of palindrome numbers in [M, N].

Sample Test Cases

Sample Input 1	Sample Output 1
`1 50`	`13`

Explanation: Single digits 1–9 (9 palindromes) + 11, 22, 33, 44 (4 palindromes) = 13 total.

Sample Input 2	Sample Output 2
`100 200`	`10`

Explanation: 101, 111, 121, 131, 141, 151, 161, 171, 181, 191 — 10 palindromes.

Approach / Hint

For each number n in [M, N]: convert n to string s; check if s == s reversed (s[::-1] in Python). Increment counter when true.

Q14	Joseph and Digital Logic — Toggle Bits Section A · Bit Manipulation	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Bit Manipulation	100 Points	90 min (Shared for both questions)

Problem Scenario

Joseph is studying digital electronics. His teacher's challenge: convert a number to binary, flip every bit (0 becomes 1, 1 becomes 0), then convert back to decimal.

Problem Statement

Given a positive integer N, find its binary representation, toggle every bit (including the MSB), and print the resulting decimal value.

Constraints

- `1 <= N <= 1000000`

Input Format

A single integer N

Output Format

Print the decimal result after toggling all bits.

Sample Test Cases

Sample Input 1	Sample Output 1
10	5

Explanation: 10 in binary = 1010. Toggle all bits → 0101 = 5.

Sample Input 2	Sample Output 2
7	0

Explanation: 7 in binary = 111. Toggle all bits → 000 = 0.

Approach / Hint

Find the bit length of N: L = floor(log2(N)) + 1. Create mask = (1 << L) - 1 (all L bits set to 1). Result = N XOR mask. This flips exactly the bits that N uses.

Q15	Perfect Donation Amount Section A · Number Theory / Divisors	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Number Theory / Divisors	100 Points	90 min (Shared for both questions)

Problem Scenario

An NGO accepts only 'perfect' donation amounts — numbers equal to the sum of all their proper divisors. Given an amount, check if it qualifies.

Problem Statement

A perfect number equals the sum of all its proper divisors (all positive divisors excluding the number itself). Given N, print YES if perfect, else NO.

Constraints

- `1 <= N <= 100000`

Input Format

A single integer N

Output Format

Print YES if N is a perfect number, else NO.

Sample Test Cases

Sample Input 1	Sample Output 1
28	YES

Explanation: Proper divisors of 28: 1, 2, 4, 7, 14. Sum = 28. Perfect!

Sample Input 2	Sample Output 2
12	NO

Explanation: Proper divisors of 12: 1, 2, 3, 4, 6. Sum = 16, not 12.

Approach / Hint

Loop i from 1 to sqrt(N): if N%i==0, add both i and N/i to sum (but do not add N itself; and don't double-count when i == N/i). Finally compare sum to N. O(sqrt N).

Q16	Prime Number Detector Section A · Number Theory / Primes	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Number Theory / Primes	100 Points	90 min (Shared for both questions)

Problem Scenario

A cryptography system uses prime numbers as encryption keys. Quickly verify whether a given number qualifies as a valid prime key.

Problem Statement

Given an integer N, determine if it is a prime number. Print Prime if it is, otherwise print Not Prime.

Constraints

- `1 <= N <= 10000000`

Input Format

A single integer N

Output Format

Print Prime or Not Prime on a single line.

Sample Test Cases

Sample Input 1	Sample Output 1
`17`	`Prime`

Explanation: 17 has no divisors other than 1 and 17 itself.

Sample Input 2	Sample Output 2
`15`	`Not Prime`

Explanation: 15 = 3 × 5; it has divisors other than 1 and itself.

Approach / Hint

Check divisibility from 2 to sqrt(N). If any i divides N evenly, it is Not Prime. Special cases: N < 2 is Not Prime; N == 2 is Prime. O(sqrt N) — never iterate all the way to N.

Q17	Fibonacci Code — N-th Term Section A · Series / Iterative DP	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Series / Iterative DP	100 Points	90 min (Shared for both questions)

Problem Scenario

A biologist models population growth using the Fibonacci sequence. She needs the N-th Fibonacci number quickly for large N. Recursive solutions are too slow.

Problem Statement

Given an integer N, print the N-th Fibonacci number (1-indexed: F(1)=1, F(2)=1, F(3)=2, ...). You MUST use an iterative approach — a naive recursive solution will exceed the time limit.

Constraints

- `1 <= N <= 10000`

Input Format

A single integer N

Output Format

Print the N-th Fibonacci number on a single line.

Sample Test Cases

Sample Input 1	Sample Output 1
7	13

Explanation: Sequence: 1, 1, 2, 3, 5, 8, 13. The 7th term is 13.

Sample Input 2	Sample Output 2
10	55

Explanation: F(10) = 55.

Approach / Hint

> Iterative: prev=1, curr=1. For i from 3 to N: next = prev + curr; prev = curr; curr = next. Return curr. O(n) time, O(1) space. Never write recursive Fibonacci for large N.

Q18	Sum of Integers in Range Section A · Math / Arithmetic Formula	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Math / Arithmetic Formula	100 Points	90 min (Shared for both questions)

Problem Scenario

A billing system calculates total service charges for all account numbers from M to N (inclusive). Each account pays a charge equal to its account number.

Problem Statement

Given two integers M and N (M <= N), compute the sum of all integers from M to N inclusive. Use the arithmetic series formula for an O(1) solution.

Constraints

- `0 <= M <= N <= 10000`

Input Format

A single line with two space-separated integers M and N

Output Format

Print the sum of all integers from M to N inclusive.

Sample Test Cases

Sample Input 1	Sample Output 1
`1 10`	`55`

Explanation: Formula: (10-1+1)*(1+10)/2 = 10*11/2 = 55.

Sample Input 2	Sample Output 2
`5 8`	`26`

Explanation: 5+6+7+8 = 26.

Approach / Hint

> One-line formula: Sum = (N - M + 1) * (M + N) / 2. This is the arithmetic progression sum formula. No loop needed — O(1).

Q19	Jack Counts Sundays Section A · Math / Modular Arithmetic	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Math / Modular Arithmetic	100 Points	90 min (Shared for both questions)

Problem Scenario

Jack is planning his lazy Sundays for a given month. He knows the starting day and the total number of days. Help him count how many Sundays he gets.

Problem Statement

Given the starting day of a month as a string (Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday) and the number of days N in the month, count the total Sundays in the month.

Constraints

- `28 <= N <= 31`
- `Starting day is exactly one of the 7 valid English day names`

Input Format

Line 1: the starting day name (e.g., Monday)

Line 2: integer N (number of days in the month)

Output Format

Print the count of Sundays as a single integer.

Sample Test Cases

Sample Input 1	Sample Output 1
`Sunday` `30`	`5`

Explanation: Sundays fall on days 1, 8, 15, 22, 29. Count = 5.

Sample Input 2	Sample Output 2
`Monday` `28`	`4`

Explanation: Sundays fall on days 7, 14, 21, 28. Count = 4.

Approach / Hint

Map day names to indices 0–6 (e.g. Monday=0, ..., Sunday=6). For each day d from 0 to N-1: if (startIndex + d) % 7 == 6, it is a Sunday. Count those days.

Q20	Speed Calculator Section A · Math / Unit Conversion	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Math / Unit Conversion	100 Points	90 min (Shared for both questions)

Problem Scenario

A physics experiment measures the time taken for a ball to travel exactly 1000 metres. Calculate and report the speed in km/hr.

Problem Statement

Given a time T in seconds, calculate the speed for a distance of 1000 metres in km/hr. Formula: speed = (1000 / T) * 3.6. Print the result rounded to exactly 2 decimal places.

Constraints

- `1 <= T <= 10000`
- `T is a positive integer`

Input Format

A single integer T (time in seconds)

Output Format

Print the speed in km/hr rounded to 2 decimal places (e.g., 360.00).

Sample Test Cases

Sample Input 1	Sample Output 1
`10`	`360.00`

Explanation: speed = (1000/10) * 3.6 = 100 * 3.6 = 360.00 km/hr.

Sample Input 2	Sample Output 2
`3600`	`1.00`

Explanation: speed = (1000/3600) * 3.6 ≈ 1.00 km/hr.

Approach / Hint

Use floating-point division: speed = (1000.0 / T) * 3.6. Print with exactly 2 decimal places. Python: print(f'{speed:.2f}'). Java: System.out.printf('%.2f', speed). C: printf('%.2f', speed).

Q21	Bubble Sort — Ascending Order Section A · Sorting / Implementation	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Sorting / Implementation	100 Points	90 min (Shared for both questions)

Problem Scenario

A sports coach needs to rank players by score. Implement sorting from scratch — use of any built-in sort function is strictly prohibited by the judge.

Problem Statement

Given an array of N integers, sort it in ascending order using Bubble Sort. You must NOT call sort() or any equivalent built-in function.

Constraints

- `1 <= N <= 1000`
- `-10000 <= arr[i] <= 10000`

Input Format

Line 1: integer N

Line 2: N space-separated integers

Output Format

Print the sorted array as N space-separated integers on a single line.

Sample Test Cases

Sample Input 1	Sample Output 1
5 64 34 25 12 22	12 22 25 34 64

Explanation: Bubble Sort repeatedly swaps adjacent out-of-order elements until fully sorted.

Sample Input 2	Sample Output 2
4 5 1 4 2	1 2 4 5

Approach / Hint

Outer loop i from 0 to N-1; inner loop j from 0 to N-i-2. Swap arr[j] and arr[j+1] if arr[j] > arr[j+1]. Optimisation: add a 'swapped' flag — if no swaps occur in a full pass, the array is already sorted; break early.

Q22	Jump Game — Reach the Last Pad Section A · Arrays / Greedy	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Greedy	100 Points	90 min (Shared for both questions)

Problem Scenario

A frog is sitting on lily pad 0 in a pond. Each lily pad shows the maximum number of hops the frog can take from that pad. Can the frog reach the last lily pad?

Problem Statement

Given an array of N non-negative integers where arr[i] is the maximum jumps from index i, determine if you can reach index N-1 starting from index 0. Print YES or NO.

Constraints

- `1 <= N <= 100000`
- `0 <= arr[i] <= 100000`

Input Format

Line 1: integer N

Line 2: N space-separated integers

Output Format

Print YES if the last index is reachable, else NO.

Sample Test Cases

Sample Input 1	Sample Output 1
6 2 3 1 1 4 0	YES

Explanation: From 0 (jump 2) → index 2 (jump 1) → index 3 (jump 1) → index 4 (jump 4) → reaches index 5.

Sample Input 2	Sample Output 2
5 3 2 1 0 4	NO

Explanation: Index 3 always has value 0 and acts as a permanent block before index 4.

Approach / Hint

Greedy: maintain maxReach (the farthest index reachable so far) = 0. For each index i: if i > maxReach, return NO (stuck). Update maxReach = max(maxReach, i + arr[i]). If at any point maxReach >= N-1, return YES.

Q23	Inventory Manager — Word Frequency Section A · HashMap / Strings	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
HashMap / Strings	100 Points	90 min (Shared for both questions)

Problem Scenario

A warehouse manager receives a shipment report as a line of product names. She needs a frequency count of each product. Any report containing digits must be flagged as invalid.

Problem Statement

Given a string of space-separated words, count and print the frequency of each unique word in their first-appearance order. If the input string contains any digit (0–9), print Invalid Input.

Constraints

- `1 <= total words <= 1000`
- `Each word contains only lowercase letters when valid`

Input Format

A single line of space-separated words

Output Format

If input has digits: print Invalid Input. Otherwise print each word and its count in first-appearance order, one per line, formatted as: word: count

Sample Test Cases

Sample Input 1	Sample Output 1
`apple mango apple banana mango apple`	`apple: 3mango: 2banana: 1`

Explanation: Frequencies printed in order of first appearance.

Sample Input 2	Sample Output 2
`box1 pen box`	`Invalid Input`

Explanation: Input contains the digit '1'.

Approach / Hint

First, scan the entire input for any digit character — if found, print Invalid Input and stop. Otherwise, use an OrderedDict (Python) or LinkedHashMap (Java) to map each word to its count while preserving insertion order.

Q24	Weekly Exercise Tracker Section A · Arrays / Floating Point	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Floating Point	100 Points	90 min (Shared for both questions)

Problem Scenario

Ananya logs her daily exercise duration in minutes for 7 consecutive days. She wants a weekly summary showing both total minutes and average daily duration.

Problem Statement

Given exactly 7 integers representing daily exercise durations in minutes, compute and print: (1) total minutes, (2) average daily duration rounded to 2 decimal places.

Constraints

- `1 <= duration[i] <= 1440`
- `Exactly 7 integers will be given`

Input Format

A single line with exactly 7 space-separated integers

Output Format

Line 1: Total: X minutes

Line 2: Average: Y minutes/day (Y rounded to 2 decimal places)

Sample Test Cases

Sample Input 1	Sample Output 1
`30 45 60 20 50 40 35`	`Total: 280 minutesAverage: 40.00 minutes/day`

Explanation: Sum = 280; Average = 280 / 7 = 40.00.

Sample Input 2	Sample Output 2
`60 60 60 60 60 60 61`	`Total: 421 minutesAverage: 60.14 minutes/day`

Explanation: 421 / 7 = 60.142857... rounds to 60.14.

Approach / Hint

Sum all 7 values (integer). Divide by 7.0 (float). Format output with exactly 2 decimal places. Watch out: 280/7 = 40.0 must be printed as 40.00, not 40.0.

Q25	All Permutations of a String Section A · Recursion / Backtracking	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
Recursion / Backtracking	100 Points	90 min (Shared for both questions)

Problem Scenario

A cryptographer needs all possible rearrangements of a secret code word to test a decryption algorithm. Print them all in lexicographic order.

Problem Statement

Given a string S of distinct lowercase letters, print all permutations of S sorted in lexicographic (alphabetical) order, one per line.

Constraints

- `1 <= |S| <= 8`
- `S contains only distinct lowercase letters`

Input Format

A single line containing string S

Output Format

All permutations, one per line, in lexicographic order.

Sample Test Cases

Sample Input 1	Sample Output 1
`abc`	`abcacbbacbcacabcba`

Explanation: All 3! = 6 permutations of 'abc' in sorted order.

Sample Input 2	Sample Output 2
`ab`	`abba`

Explanation: 2 permutations.

Approach / Hint

Sort the string first. Use backtracking: swap characters to fix one at each position and recurse for the rest. Collect all results in a list, sort it, then print each. For strings up to length 8, this runs in at most 8! = 40,320 operations.

SECTION B — PREDICTED QUESTIONS

Questions 26 to 37 · High-probability topics based on 2024–2025 trend analysis

Q26	Sliding Window Maximum Section B · Arrays / Deque / Sliding Window	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Deque / Sliding Window	100 Points	90 min (Shared for both questions)

Problem Scenario

A weather station records hourly temperatures. A scientist wants the peak temperature for every consecutive window of K hours throughout the day.

Problem Statement

Given an array of N integers and window size K, print the maximum element for every contiguous subarray of size K. There are N-K+1 windows in total.

Constraints

- `1 <= K <= N <= 100000`
- `-10000 <= arr[i] <= 10000`

Input Format

Line 1: two integers N and K separated by a space

Line 2: N space-separated integers

Output Format

Print N-K+1 space-separated maximum values on a single line.

Sample Test Cases

Sample Input 1	Sample Output 1
8 3 1 3 -1 -3 5 3 6 7	3 3 5 5 6 7

Explanation: Windows of 3: [1,3,-1]→3, [3,-1,-3]→3, [-1,-3,5]→5, [-3,5,3]→5, [5,3,6]→6, [3,6,7]→7.

Sample Input 2	Sample Output 2
5 2 4 3 2 1 5	4 3 2 5

Approach / Hint

Use a deque of indices (decreasing order of values). For each element i: remove indices outside the window (< i-K+1) from front. Remove indices from back while arr[deque.back()] <= arr[i]. Add i to back. After processing first K elements, record arr[deque.front()] for each window.

Q27	Detect Cycle in Linked List Section B · Linked List / Floyd's Algorithm	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
Linked List / Floyd's Algorithm	100 Points	90 min (Shared for both questions)

Problem Scenario

A network engineer suspects a data packet is caught in a routing loop. Given the routing path as a linked list, determine whether a cycle exists.

Problem Statement

You are given N node values and an integer P. The list is a singly linked list; the tail node points back to the node at index P (0-indexed), or has no next pointer if P = -1. Determine if a cycle exists. Print Cycle Detected or No Cycle.

Constraints

- `1 <= N <= 100000`
- `-1 <= P < N`

Input Format

Line 1: integer N

Line 2: N space-separated node values

Line 3: integer P (tail connects to index P; -1 means no cycle)

Output Format

Print Cycle Detected or No Cycle.

Sample Test Cases

Sample Input 1	Sample Output 1
`51 2 3 4 52`	`Cycle Detected`

Explanation: Tail node (5) connects back to index 2 (value 3), creating a cycle.

Sample Input 2	Sample Output 2
`41 2 3 4-1`	`No Cycle`

Explanation: Linear list with no back-pointer.

Approach / Hint

Floyd's Tortoise and Hare: slow = head, fast = head. While fast and fast.next are not null: slow = slow.next; fast = fast.next.next. If slow == fast, a cycle exists.

Q28	Longest Common Subsequence Section B · Dynamic Programming	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
Dynamic Programming	100 Points	90 min (Shared for both questions)

Problem Scenario

Two students compared their study schedules (represented as strings). Find the length of the longest sequence of subject codes they both follow, in the same relative order.

Problem Statement

Given two strings A and B, find the length of their Longest Common Subsequence (LCS). A subsequence preserves relative order but does not need to be contiguous.

Constraints

- `1 <= |A|, |B| <= 1000`
- `Strings contain only lowercase letters`

Input Format

Line 1: string A

Line 2: string B

Output Format

Print the length of the LCS as a single integer.

Sample Test Cases

Sample Input 1	Sample Output 1
abcde ace	3

Explanation: LCS is 'ace' (length 3).

Sample Input 2	Sample Output 2
abc abc	3

Explanation: LCS is 'abc' itself.

Approach / Hint

2D DP table dp[i][j] = LCS length of A[0..i-1] and B[0..j-1]. If A[i-1] == B[j-1]: dp[i][j] = dp[i-1][j-1] + 1. Else: dp[i][j] = max(dp[i-1][j], dp[i][j-1]). Answer = dp[|A|][|B|].

Q29	Caesar Cipher Encryption Section B · Strings / ASCII Arithmetic	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Strings / ASCII Arithmetic	100 Points	90 min (Shared for both questions)

Problem Scenario

A spy agency encodes messages by shifting every letter K positions forward in the alphabet (wrapping Z back to A). Encode the given message.

Problem Statement

Given a string S of uppercase English letters and an integer K, encrypt S using Caesar Cipher: shift each character K positions forward, wrapping from Z back to A. Print the encrypted string.

Constraints

- `1 <= |S| <= 100000`
- `S contains only uppercase English letters A-Z`
- `1 <= K <= 25`

Input Format

Line 1: string S

Line 2: integer K

Output Format

Print the encrypted string (uppercase).

Sample Test Cases

Sample Input 1	Sample Output 1
HELLO 3	KHOOR

Explanation: H→K, E→H, L→O, L→O, O→R.

Sample Input 2	Sample Output 2
XYZ	ABC

3	

Explanation: X→A, Y→B, Z→C (wrap-around applies).

Approach / Hint

For each character c: encrypted_char = chr((ord(c) - ord('A') + K) % 26 + ord('A')). The modulo 26 handles the wrap-around from Z back to A automatically.

Q30	Matrix Spiral Traversal Section B · 2D Arrays / Simulation	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
2D Arrays / Simulation	100 Points	90 min (Shared for both questions)

Problem Scenario

A printer needs to print all elements of a grid in clockwise spiral order — starting from the top-left corner and spiralling inward.

Problem Statement

Given an M × N matrix of integers, print all elements in clockwise spiral order on a single line separated by spaces.

Constraints

- `1 <= M, N <= 100`
- `-1000 <= matrix[i][j] <= 1000`

Input Format

Line 1: two integers M and N separated by a space

Next M lines: N space-separated integers each

Output Format

Print all elements in clockwise spiral order on one line.

Sample Test Cases

Sample Input 1	Sample Output 1
`3 31 2 34 5 67 8 9`	`1 2 3 6 9 8 7 4 5`

Explanation: Top row L→R, right column T→B, bottom row R→L, left column B→T, then inner.

Sample Input 2	Sample Output 2
`2 41 2 3 45 6 7 8`	`1 2 3 4 8 7 6 5`

Approach / Hint

Four boundary pointers: top=0, bottom=M-1, left=0, right=N-1. In a while loop: traverse top row (left→right, then top++); right column (top→bottom, then right--); bottom row (right→left, then bottom--); left column (bottom→top, then left++). Repeat while top <= bottom and left <= right.

Q31	Trapping Rain Water Section B · Arrays / Two Pointer	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Two Pointer	100 Points	90 min (Shared for both questions)

Problem Scenario

After heavy rain, water collects between buildings of different heights. Given the building heights, calculate the total volume of rainwater trapped between them.

Problem Statement

Given an array of N non-negative integers representing building heights, compute the total units of rainwater trapped after rain fills all gaps.

Constraints

- `1 <= N <= 100000`
- `0 <= height[i] <= 10000`

Input Format

Line 1: integer N

Line 2: N space-separated integers

Output Format

Print a single integer: total units of water trapped.

Sample Test Cases

Sample Input 1	Sample Output 1
12 0 1 0 2 1 0 1 3 2 1 2 1	6

Explanation: 6 units of water are trapped in the valleys between the buildings.

Sample Input 2	Sample Output 2
6 4 2 0 3 2 5	9

Explanation: Total trapped water = 9 units.

Approach / Hint

Two-pointer: left=0, right=N-1, leftMax=0, rightMax=0, result=0. While left < right: if height[left] <= height[right]: if height[left] >= leftMax update leftMax else add leftMax-height[left] to result; left++. Else mirror logic from right. O(n), O(1).

Q32	Maximum Subarray Sum — Kadane's Algorithm Section B · Arrays / Greedy / DP	DIFFICULTY Easy-Medium

TOPIC	MAX SCORE	TIME LIMIT
Arrays / Greedy / DP	100 Points	90 min (Shared for both questions)

Problem Scenario

A stock trader wants to find the most profitable consecutive trading window. Given daily profit/loss values, find the maximum sum of any contiguous trading period.

Problem Statement

Given an array of N integers (positive and/or negative), find the contiguous subarray with the largest sum and print that sum.

Constraints

- `1 <= N <= 100000`
- `-10000 <= arr[i] <= 10000`

Input Format

Line 1: integer N

Line 2: N space-separated integers

Output Format

Print the maximum subarray sum as a single integer.

Sample Test Cases

Sample Input 1	Sample Output 1
9 -2 1 -3 4 -1 2 1 -5 4	6

Explanation: Subarray [4, -1, 2, 1] has the maximum sum = 6.

Sample Input 2	Sample Output 2
5 1 2 3 4 5	15

Explanation: The entire array has the maximum sum.

Approach / Hint

Kadane's algorithm: maxSoFar = maxEndingHere = arr[0]. For i from 1 to N-1: maxEndingHere = max(arr[i], maxEndingHere + arr[i]); maxSoFar = max(maxSoFar, maxEndingHere). Return maxSoFar.

Q33	Valid Parentheses / Balanced Brackets Section B · Stack / String	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Stack / String	100 Points	90 min (Shared for both questions)

Problem Scenario

A compiler must verify that all brackets in source code are properly opened and closed in the correct nesting order. Build the bracket validator.

Problem Statement

Given a string S containing only the characters (,), {, }, [,], determine if the brackets are balanced and properly nested. Print Valid or Invalid.

Constraints

- `1 <= |S| <= 100000`
- `S contains only ( ) { } [ ]`

Input Format

A single line containing the bracket string S

Output Format

Print Valid if brackets are balanced, else Invalid.

Sample Test Cases

Sample Input 1	Sample Output 1
`()[]{}`	`Valid`

Explanation: Each open bracket is closed by the matching closing bracket in correct order.

Sample Input 2	Sample Output 2
`([)]`	`Invalid`

Explanation: '[' is closed by ')' before it is closed by ']' — incorrect nesting.

Approach / Hint

Stack approach: for each character c: if c is an open bracket push it. If c is a closing bracket: if stack is empty or top of stack doesn't match, return Invalid; else pop the top. At end: if stack is empty return Valid, else Invalid.

Q34	GCD of Array Elements Section B · Math / Euclidean Algorithm	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Math / Euclidean Algorithm	100 Points	90 min (Shared for both questions)

Problem Scenario

A factory has N machines that complete their cycles at different intervals (in seconds). The GCD of all cycle times tells the factory how often all machines sync up simultaneously.

Problem Statement

Given an array of N positive integers, find the GCD (Greatest Common Divisor) of all elements by applying the Euclidean algorithm iteratively.

Constraints

- `1 <= N <= 100000`
- `1 <= arr[i] <= 1000000`

Input Format

Line 1: integer N

Line 2: N space-separated integers

Output Format

Print the GCD of all array elements as a single integer.

Sample Test Cases

Sample Input 1	Sample Output 1
4 12 18 24 36	6

Explanation: GCD(12,18)=6; GCD(6,24)=6; GCD(6,36)=6.

Sample Input 2	Sample Output 2
3 7 14 21	7

Explanation: All three are divisible by 7.

Approach / Hint

Euclidean GCD: gcd(a, b) = gcd(b, a % b) until b == 0. Apply iteratively: result = arr[0]; for i from 1 to N-1: result = gcd(result, arr[i]). The final result is the GCD of all elements.

Q35	Coin Change — Minimum Coins Section B · Dynamic Programming	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
Dynamic Programming	100 Points	90 min (Shared for both questions)

Problem Scenario

An ATM holds coins of certain denominations. A customer wants to withdraw exactly Rs. T. Find the minimum number of coins needed, or report that it is impossible.

Problem Statement

Given an array of coin denominations and a target amount T, find the minimum number of coins needed to make exactly T. If it is impossible, print -1.

Constraints

- `1 <= number of denominations <= 12`
- `1 <= coin[i] <= 1000`
- `1 <= T <= 10000`

Input Format

Line 1: integer N (number of coin types)

Line 2: N space-separated coin denominations

Line 3: integer T (target amount)

Output Format

Print the minimum number of coins to make T, or -1 if impossible.

Sample Test Cases

Sample Input 1	Sample Output 1
`31 5 611`	`2`

Explanation: 6 + 5 = 11 using 2 coins (optimal). Using 1s alone would require 11 coins.

Sample Input 2	Sample Output 2
`32 5 103`	`-1`

Explanation: It is impossible to make 3 using only {2, 5, 10}.

Approach / Hint

1D DP: dp[0] = 0; dp[i] = infinity for i > 0. For each amount i from 1 to T: for each coin c <= i: dp[i] = min(dp[i], dp[i - c] + 1). If dp[T] is still infinity after the loop, print -1.

Q36	Bubble Sort — Descending Order Section B · Sorting / Implementation	DIFFICULTY Easy

TOPIC	MAX SCORE	TIME LIMIT
Sorting / Implementation	100 Points	90 min (Shared for both questions)

Problem Scenario

A leaderboard system needs players sorted by score from highest to lowest. Implement sorting manually — built-in sort functions are strictly not allowed.

Problem Statement

Given an array of N integers, sort it in DESCENDING order using Bubble Sort. You must NOT use any built-in sorting function.

Constraints

- `1 <= N <= 1000`
- `0 <= arr[i] <= 100000`

Input Format

Line 1: integer N

Line 2: N space-separated integers

Output Format

Print the array sorted in descending order as N space-separated integers.

Sample Test Cases

Sample Input 1	Sample Output 1
5 10 40 30 20 50	50 40 30 20 10

Explanation: Sorted highest to lowest.

Sample Input 2	Sample Output 2
4 5 5 3 1	5 5 3 1

Explanation: Already in descending order.

Approach / Hint

Same as ascending Bubble Sort but change the swap condition: swap arr[j] and arr[j+1] when arr[j] < arr[j+1] (instead of >). This bubbles the largest values to the front.

Q37	BFS — Minimum Steps in Grid Section B · Graphs / Breadth-First Search	DIFFICULTY Medium

TOPIC	MAX SCORE	TIME LIMIT
Graphs / Breadth-First Search	100 Points	90 min (Shared for both questions)

Problem Scenario

A robot placed at the top-left corner of a grid must reach the bottom-right corner. Some cells are blocked. Moving through one open cell costs one step. Find the minimum steps required.

Problem Statement

Given an M×N grid where 0 = open cell and 1 = blocked cell, find the minimum number of steps from cell (0,0) to cell (M-1, N-1) using BFS. Movement is allowed in 4 directions (Up, Down, Left, Right). If no path exists, print -1.

Constraints

- `1 <= M, N <= 100`
- `Grid contains only 0s and 1s`
- `Start (0,0) and end (M-1,N-1) are guaranteed to be 0`

Input Format

Line 1: two integers M and N separated by a space

Next M lines: N space-separated integers (0 or 1)

Output Format

Print the minimum number of steps, or -1 if unreachable.

Sample Test Cases

Sample Input 1	Sample Output 1
`3 30 0 01 1 01 1 0`	`4`

Explanation: Path: (0,0)→(0,1)→(0,2)→(1,2)→(2,2) = 4 steps.

Sample Input 2	Sample Output 2
`2 20 11 0`	`-1`

Explanation: All possible paths are blocked.

Approach / Hint

BFS from (0,0): use a queue with entries (row, col, steps). Mark cells visited when enqueued. For each cell dequeued, check all 4 neighbours: if valid (in bounds, not blocked, not visited), enqueue with steps+1. Return steps when (M-1, N-1) is reached; return -1 if queue empties.